Name: Yash Gawade
Roll no: 46

```
from google.colab import files

uploaded = files.upload()
```

Choose files Iris.csv
Iris.csv(text/csv) - 5107 bytes, last modified: 04/02/2026 - 100% done
Saving Iris.csv to Iris.csv

```
import pandas as pd

df = pd.read_csv("Iris.csv")
```

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style ="whitegrid")
plt.rcParams["figure.figsize"] = (8,5)

df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: Generate code with df New interactive sheet

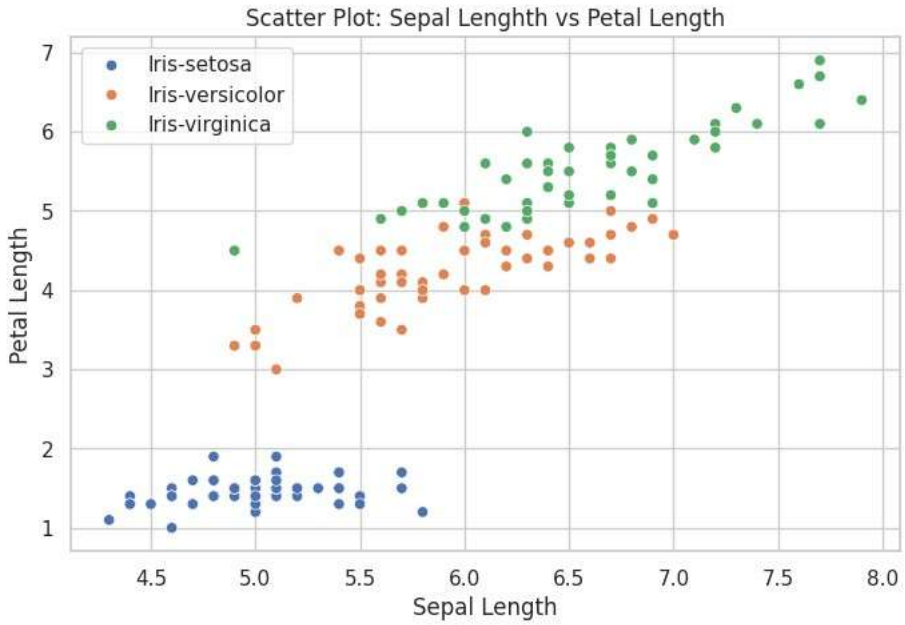
```
df.info()
df.describe()
df["Species"].value_counts()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

Species	count
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

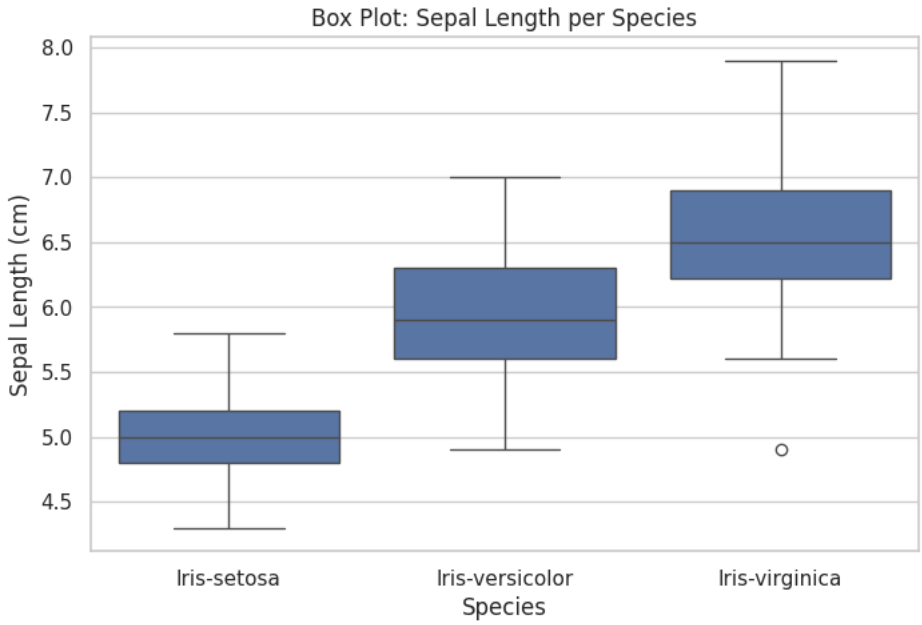
dtype: int64

```
plt.figure()
sns.scatterplot(
    data = df,
    x = "SepalLengthCm",
    y = "PetalLengthCm",
    hue = "Species"
)

plt.title("Scatter Plot: Sepal Lenghth vs Petal Length")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.legend()
plt.show()
```

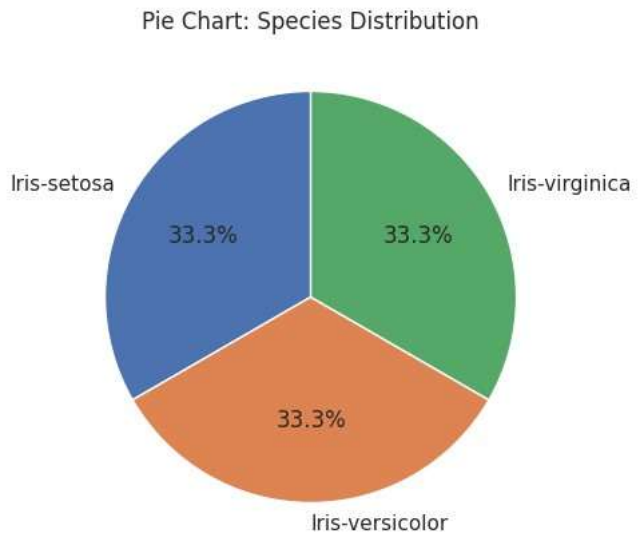



```
plt.figure()
sns.barplot(
    data = df,
    x = "Species",
    y = "SepalLengthCm",
    estimator = "mean",
    errorbar = "sd"
)

plt.title("Bar Plot: Mean Petal Length per Species")
plt.xlabel("Species")
plt.ylabel("Mean Petal Length (cm)")
plt.show()
```

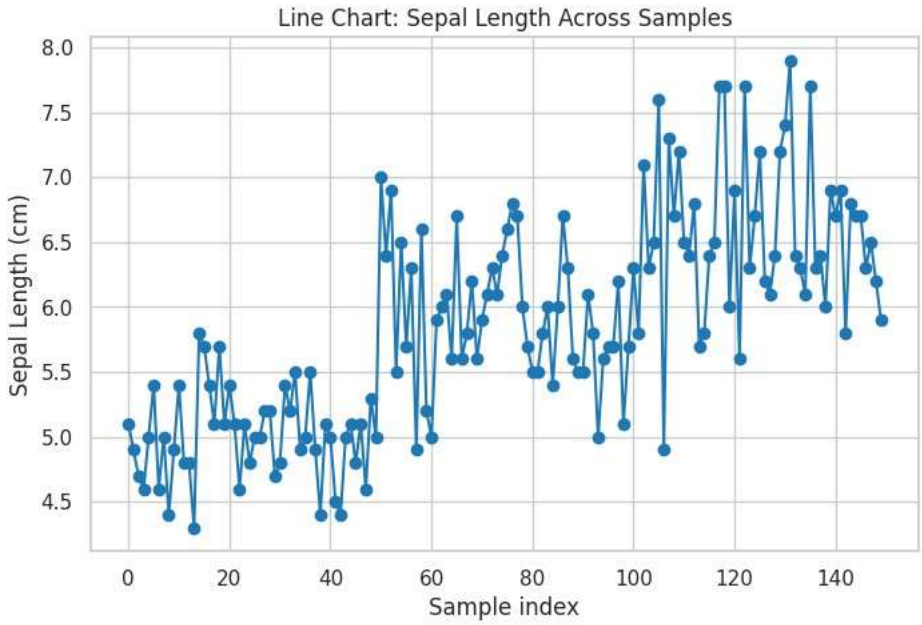
```
plt.figure()
sns.boxplot(
    data = df,
    x = "Species",
    y = "SepalLengthCm"
)

plt.title("Box Plot: Sepal Length per Species")
plt.xlabel("Species")
plt.ylabel("Sepal Length (cm)")
plt.show()
```

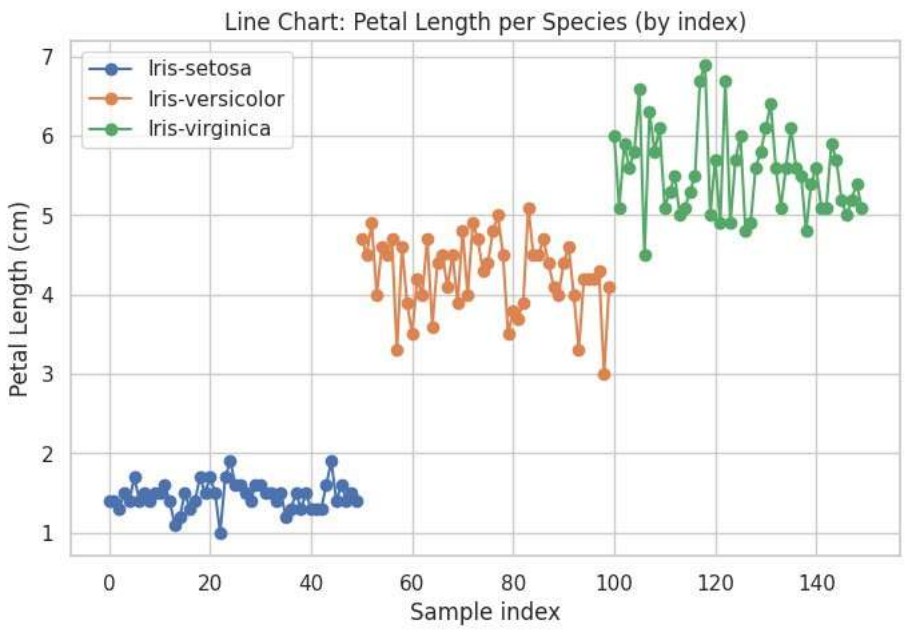



```
species_counts = df["Species"].value_counts()

plt.figure()
plt.pie(
    species_counts.values,
    labels = species_counts.index,
    autopct = "%1.1f%%",
    startangle = 90
)
plt.title("Pie Chart: Species Distribution")
plt.show()
```




```
# If your CSV has an Id column, you can use it; otherwise use the index
x = df.index  # or df["Id"] if exists

plt.figure()
plt.plot(
    x,
    df["SepalLengthCm"],
    marker="o",
    linestyle="-",
    color="tab:blue"
)
plt.title("Line Chart: Sepal Length Across Samples")
plt.xlabel("Sample index")
plt.ylabel("Sepal Length (cm)")
plt.show()
```




```
plt.figure()
for species, subdf in df.groupby("Species"):
    plt.plot(
        subdf.index,
        subdf["PetalLengthCm"],
        marker="o",
        linestyle="-",
        label=species
    )

plt.title("Line Chart: Petal Length per Species (by index)")
plt.xlabel("Sample index")
plt.ylabel("Petal Length (cm)")
plt.legend()
plt.show()
```